The t_c-Cederbaum model: ghost Gutzwiller by direct minimization

(core fragment with $B_c = 1$, 3 and 5 quasiparticle orbitals)

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This note defines the Hamiltonian that is solved, the t_c-Cederbaum model (the Cederbaum model of core-hole satellites plus a weak core–virtual hopping t_c; $t_c = 0$ is the original Cederbaum model), its exact solution by diagonalization (ED), the ghost-Gutzwiller (gGut) variational problem used for it, the direct minimization that solves that problem, and the comparison of the gGut results with ED, for a core fragment without ghosts ($B_c = 1$) and with two and four ghosts ($B_c = 3$, 5). The aim of the test is to see whether gGut reproduces the satellite structure of the core-hole spectrum of this model. Quantities are per spin unless stated otherwise; solutions are real and paramagnetic.

1 The Hamiltonian

Three orbitals, c (core), o (occupied) and v (virtual), each with two spins. The t_c-Cederbaum model is $H_{\mathrm{tot}} = H + H_t$, where H is the Cederbaum model (the o–v hopping conditioned on a core hole) and H_t the added hopping:

$$\begin{aligned} H_{\mathrm{tot}} = {} & g \sum_{\sigma} (n_{v\sigma} - n_{o\sigma}) - E_c \sum_{\sigma} n_{c\sigma} - V \sum_{\sigma\sigma'} (1 - n_{c\sigma}) \left(c^{\dagger}_{v\sigma'} c_{o\sigma'} + \mathrm{h.c.} \right) \\ & + t_c \sum_{\sigma} \left(c^{\dagger}_{c\sigma} c_{v\sigma} + \mathrm{h.c.} \right), \end{aligned} \tag{1}$$

with $E_c > 0$, orbital gap g, core–valence coupling V, and a weak core–virtual hopping t_c. The prefactor $\sum_{\sigma}(1 - n_{c\sigma}) = 2 - n_c$ counts the core holes, so the o–v hopping acts only when the core is not full. The particle number and S_z are conserved. The core occupation n_c is conserved only at $t_c = 0$. Energies are measured in units of g ($g = 1$), so the parameters are the ratios E_c/g, V/g and t_c/g. All results below use $E_c/g = 10$, $V/g \in \{0.5, 1, 2\}$ and $t_c/g \leq 0.5$; the ground state then lies in the sector with two electrons of each spin.

2 Exact diagonalization

Diagonalization. H_{tot} is diagonalized exactly in the 64-dimensional Fock space of six spin orbitals (Jordan–Wigner operators). At $t_c = 0$ the ground state is the product state $|220\rangle$ (notation $|n_c n_o n_v\rangle$: core and o doubly occupied, v empty) with $E = -2E_c - 2g$ and core occupation $\nu_c = \langle n_c \rangle = 2$; for $t_c > 0$ it acquires a core-hole component and $\nu_c < 2$. The core-hole Green's function for spin $\uparrow$ is

$$G^{h}_{cc}(\omega) = \sum_{n} \frac{|\langle n | c_{c\uparrow} | \mathrm{GS} \rangle|^2}{\omega + (E_n - E_{\mathrm{GS}}) - i\eta}, \tag{2}$$

with poles at $\omega = -E_i$ and weights p_i. At $t_c = 0$ there are three poles; with $\alpha = V/g$,

$$\begin{aligned} E_1 &= E_c + 2g - 2g\sqrt{1+\alpha^2}, \quad p_1 = \frac{(1+\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \\ E_2 &= E_c + 2g, \qquad\qquad\qquad\quad p_2 = \frac{\alpha^2}{2(1+\alpha^2)}, \\ E_3 &= E_c + 2g + 2g\sqrt{1+\alpha^2}, \quad p_3 = \frac{(1-\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \end{aligned} \tag{3}$$

which the ED reproduces to 9×10^{-15}. For $t_c > 0$ the poles move at $O(t_c^2)$ and the middle pole splits into two lines $\sim 10^{-4}$ apart (Tab. 2); Tab. 1 gives the ground-state quantities.

V/g	t_c/g	$(E_0 + 2E_c + 2g)/g$	$2 - \nu_c$	$\rho^{cv} = \langle c^\dagger_{c\uparrow} c_{v\uparrow}\rangle$
1	0.5	-4.57×10^{-2}	4.19×10^{-3}	-4.56×10^{-2}
1	0.2	-7.32×10^{-3}	6.74×10^{-4}	-1.83×10^{-2}
1	0.1	-1.83×10^{-3}	1.69×10^{-4}	-9.15×10^{-3}
1	0.05	-4.58×10^{-4}	4.22×10^{-5}	-4.58×10^{-3}
2	0.5	-4.67×10^{-2}	4.45×10^{-3}	-4.66×10^{-2}
2	0.05	-4.68×10^{-4}	4.48×10^{-5}	-4.68×10^{-3}

Table 1: Exact ground state of $H_{\rm tot}$: $2 - \nu_c \propto t_c^2$ and $\rho^{cv} \propto t_c$.

$V/g = 1$	$-E_1$	$-E_2$ (split pair)	$-E_3$	p_1
$t_c \to 0$ (Eq. 3)	−9.1716	−12	−14.8284	0.7286
$t_c/g = 0.5$	−9.2019	−12.0429, −12.0492	−14.8878	0.7319
$t_c/g = 0.1$	−9.1728	−12.0017, −12.0020	−14.8308	0.7287
$t_c/g = 0.05$	−9.1719	−12.0004, −12.0005	−14.8290	0.7286

Table 2: Exact hole poles of G^h_{cc} at $V/g = 1$ (weights $> 10^{-3}$); the split pair has total weight $p_2 = 0.25$.

3 The ghost-Gutzwiller variational problem

Fragments and quasiparticle space. Each orbital $I \in \{c, o, v\}$ is a fragment with B_I quasiparticle orbitals $d_{Ia\sigma}$, $a = 1, \ldots, B_I$: $B_o = B_v = 1$ (no local interaction) and $B_c \in \{1, 3, 5\}$, i.e. $B_c - 1$ ghost orbitals on the core. The Gutzwiller wave function is $|\Psi_G\rangle = \prod_I P_I |\psi_{qp}\rangle$ with $|\psi_{qp}\rangle$ a Slater determinant that fills $N_{\rm occ} = 2 + (B_c - 1)/2$ of the $B_c + 2$ levels per spin. Its one-body density matrix is $\rho^{IJ}_{ab} = \langle d^\dagger_{Ia\sigma} d_{Jb\sigma}\rangle$.

Embedding states. The projector of each fragment is represented by a real normalized state Φ_I in the Fock space of the physical orbital I plus B_I bath orbitals per spin, restricted to the half-filled sector with $(1 + B_I)/2$ electrons of each spin (B_I odd). Writing $c_{I\sigma}$ for the physical and $d_{Ia\sigma}$ for the bath orbitals, the following quantities are expectation values in Φ_I (spin averaged):

$$\begin{aligned} D_{I,a} &= \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} c_{I\sigma}\rangle, \quad \Delta^{bb}_{I,ab} = \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} d_{Ib\sigma}\rangle, \\ \nu_c &= \langle n_c\rangle, \qquad\qquad\qquad K_{\gamma\delta} = \Big\langle n_c \sum_\sigma d_{c\delta\sigma} d^\dagger_{c\gamma\sigma}\Big\rangle, \end{aligned} \tag{4}$$

and $E_I^{loc} = \langle H_I^{loc} \rangle$ with $H_c^{loc} = -E_c n_c$, $H_o^{loc} = -g n_o$, $H_v^{loc} = +g n_v$ ($n_I = \sum_\sigma n_{I\sigma}$). From them:

$$ {}^I\Delta = \mathbb{I} - \Delta_I^{bb}, \qquad S_I = \left[{}^I\Delta(\mathbb{I} - {}^I\Delta) + \varepsilon\,\mathbb{I} \right]^{1/2}, \qquad R_I = S_I^{-1} D_I, \qquad C = (\mathbb{I} - {}^c\Delta + \varepsilon\,\mathbb{I})^{-1}, \quad (5) $$

where ε is a small regularizing shift (10^{-9} unless stated) and $R_I \in \mathbb{R}^{B_I}$ is the renormalization vector, $c_{I\sigma}^\dagger \to \sum_a R_{I,a} d_{Ia\sigma}^\dagger$. The non-local core–valence correlation enters through the tensor

$$ \bar{\Theta} = \tfrac{1}{2} S_c^{-1} K S_c^{-1} - \nu_c\, C, \qquad u = \rho^{cv} R_v, \quad w = \rho^{co} R_o, \quad (6) $$

which follows from evaluating $\langle \Psi_G | n_{c\sigma} c_{o\sigma'}^\dagger c_{v\sigma'} | \Psi_G \rangle$ with the Gutzwiller approximation (derived in `main.tex`, Eqs. ThetaEff–ThetaEqn, and `SI.tex`, S2).

Energy functional. The energy is $E = \langle \Psi_G | H_{\rm tot} | \Psi_G \rangle$, evaluated term by term from Eq. (1) with $c_{I\sigma}^\dagger \to \sum_a R_{I,a} d_{Ia\sigma}^\dagger$ and Wick's theorem on $|\psi_{qp}\rangle$.

- *Local terms.* $g(n_v - n_o) - E_c n_c = \sum_I H_I^{loc}$ is evaluated in the embedding states, $\sum_I E_I^{loc}$.
- *Core–virtual hopping.* Per spin, $\langle c_{c\sigma}^\dagger c_{v\sigma} \rangle = R_c^T \rho^{cv} R_v$. The sum over the two spins and the hermitian conjugate give a factor 4: $4 t_c\, R_c^T \rho^{cv} R_v$.
- *Conditioned o–v hopping.* The weight $(2 - n_c)$ multiplies $\sum_{\sigma'} (c_{v\sigma'}^\dagger c_{o\sigma'} + \text{h.c.})$ and the expectation value does not factorize, since n_c fluctuates and is correlated with the hopping. Per spin, $\langle (2 - n_c)\, c_{v\sigma}^\dagger c_{o\sigma} \rangle = (2 - \nu_c)\, R_v^T \rho^{vo} R_o + w^T \bar{\Theta}\, u$: the first part is the mean-field one, with the average number of core holes $2 - \nu_c$; the second is the connected core–valence correlation, built from $\bar{\Theta}$, u and w defined above. Spin sum and hermitian conjugate again give a factor 4.

Adding the three contributions with the prefactors of Eq. (1):

$$ E = \underbrace{\sum_I E_I^{loc}}_{\text{local}} + \underbrace{4 t_c\, R_c^T \rho^{cv} R_v}_{t_c \text{ hopping}} - \underbrace{4V (2 - \nu_c)\, R_v^T \rho^{vo} R_o}_{o\text{–}v \text{ mean field}} - \underbrace{4V\, w^T \bar{\Theta}\, u}_{\text{connected}}. \quad (7) $$

Written out with the definitions above, the dependences are: E_I^{loc}, ν_c, Δ_I^{bb} and K are expectation values in Φ_I; $R_I = S_I^{-1} D_I$ with S_I a function of Δ_I^{bb}; $\bar{\Theta}$ a function of ν_c, K and ${}^c\Delta$; $u = \rho^{cv} R_v$ and $w = \rho^{co} R_o$. So E depends on the embedding states only through E_I^{loc}, D_I, Δ_I^{bb}, ν_c and K, and on $|\psi_{qp}\rangle$ only through ρ.

The limit $t_c = 0$. At $t_c = 0$ the core occupation is conserved and the exact ground state is the product state $|220\rangle$ of Sec. 2, with $\nu_c = 2$. There is no core–valence correlation, so the core is coherent neither with v nor with o: $\rho^{cv} = \rho^{co} = 0$ and $u = w = 0$. Term by term in Eq. (7):

- the t_c hopping vanishes through its explicit factor t_c;
- the mean-field o–v term vanishes through $2 - \nu_c = 0$;
- the connected term vanishes through $u = w = 0$.

Hence

$$ E\big|_{t_c = 0} = \sum_I E_I^{loc} = -2E_c - 2g, \quad (8) $$

which is the exact energy. Two consequences follow. First, V does not appear: every V term of Eq. (7) is zero. Second, $R_c = S_c^{-1} D_c$ appears in Eq. (7) *only* in the t_c hopping term (the other quantities, E_I^{loc}, ν_c, Δ_I^{bb}, K, do not involve D_c, and u, w contain R_v, R_o but not R_c), so with that term gone E is independent of D_c and the minimization leaves R_c undetermined. At $V = 0$ and $t_c \neq 0$ the situation is different: the two V terms drop out, but $E = \sum_I E_I^{loc} + 4t_c\, R_c^T \rho^{cv} R_v$ still depends on R_c, which is therefore determined by the hopping.

These two consequences are the reason for the hopping t_c. The quantity of interest is not the ground-state energy but the core-hole Green's function G^h_{cc}, whose three poles are final-state physics: the three $N-1$ eigenstates reached when a core electron is removed and V then moves electrons between o and v. gGut does not compute $N-1$ states; the spectrum comes from the quasiparticle Hamiltonian H_{qp} (Sec. 4), whose parameters R_I, λ_I are fixed by the ground-state variational problem. At $t_c = 0$ that problem contains no V and leaves R_c free, so the G^h_{cc} built from R_c carries no information about the satellites, although the ground state itself is exact. A weak t_c cures both defects at once: it makes R_c determined and slightly correlates the ground state ($\nu_c < 2$), so that the V terms of Eq. (7) switch on and R_c, H_{qp} acquire V-dependent values. It is a probe, not a change of the question: the exact poles move only at $O(t_c^2)$ (Sec. 2), so a method that captures the satellites should reproduce them at small t_c.

The hopping has to couple $|220\rangle$ to other states, which fixes its form. A hopping $t_c(c_c^\dagger c_o + \text{h.c.})$ is useless: both orbitals are full in $|220\rangle$, so it annihilates the state, and $|220\rangle$ remains an exact eigenstate for every t_c (checked by ED). With $c_v^\dagger c_c$, instead, a core electron moves to the empty v at an energy cost $E_c + g$, so the ground state acquires a core-hole component of amplitude $\sim t_c/(E_c+g)$ and $2 - \nu_c = O(t_c^2)$. This is exactly the combination that Eq. (7) needs to become V-dependent: the term $4t_c\, R_c^T \rho^{cv} R_v$ determines R_c, and $2 - \nu_c \neq 0$ activates the o–v terms.

Variational problem.

$$\min_{\{\Phi_I\},\ \psi_{qp}} E \qquad \text{subject to} \qquad \langle \Phi_I | \Phi_I \rangle = 1, \qquad \rho^{II} = {}^I\Delta(\Phi_I) \quad (I = c, o, v). \tag{9}$$

The second constraint ties the local quasiparticle density matrix to the embedding state; it has $B_I(B_I+1)/2$ independent components per fragment (8 for $B_c = 3$). Multipliers λ_I of this constraint define the quasiparticle Hamiltonian H_{qp} of Sec. 4.

4 Direct minimization

Problem (9) is solved directly (`not_imp_gga_cederbaum.py`), as in the direct-minimization solver used for the Hubbard dimer, without any self-consistency cycle.

Variables. The embedding states are parametrized by unnormalized real vectors x_I in the half-filled sector ($\Phi_I = x_I/|x_I|$; dimensions 36, 4, 4 for $B_c = 3$ and 400, 4, 4 for $B_c = 5$), and the Slater determinant by a real $(B_c + 2) \times N_{\text{occ}}$ matrix X with $\rho = X(X^TX)^{-1}X^T$. The constraints $\rho^{II} = {}^I\Delta(\Phi_I)$ are imposed as equalities (SLSQP, tolerance 10^{-13} on the energy). Everything is evaluated with sector-projected dense matrices of the operators above.

Gradients. The analytic gradient of Eq. (7) follows from the chain rule through $R_I(D_I, \Delta_I^{bb})$, $\bar{\Theta}(\nu_c, K, {}^c\Delta)$ and $\rho(X)$:

$$dE = \sum_I 4\, h_I \cdot dD_I + \sum_I 2\,\mathrm{Tr}[\Lambda_I\, d\Delta_I^{bb}] + \alpha\, d\nu_c + \mathrm{Tr}[\Xi\, dK] + \sum_I dE_I^{loc} + 2\,\mathrm{Tr}[H_{\text{off}}\, d\rho], \tag{10}$$

with $h_I = S_I^{-1} X_I$, where $X_I = \frac{1}{4}\,\partial E/\partial R_I$,

$$\begin{aligned} X_c &= t_c\,\rho^{cv} R_v, \qquad X_o = -V\big[(2-\nu_c)\rho^{ov} R_v + \rho^{oc}\bar{\Theta} u\big], \\ X_v &= t_c\,\rho^{vc} R_c - V\big[(2-\nu_c)\rho^{vo} R_o + \rho^{vc}\bar{\Theta} w\big], \end{aligned} \tag{11}$$

and with the matrices $\Lambda_I = -\frac{1}{2}\nabla^s_{{}^I\Delta} E_{\rm tot}$ (derivative of E with respect to ${}^I\Delta$ at fixed D_I through R_I, S_I and $\bar{\Theta}$) given by

$$\begin{aligned} \Lambda_{o,v} &= G\big({}^I\Delta,\, h_I R_I^T\big), \\ \Lambda_c &= G\big({}^c\Delta,\, h_c R_c^T - \tfrac{1}{4} Y\big) - V\nu_c\, C\,(u w^T + w u^T)\, C, \\ Y &= V\big(S_c^{-1} K\,\Pi + \Pi\, K\, S_c^{-1}\big), \quad \Pi = x y^T + y x^T, \quad x = S_c^{-1} w,\ y = S_c^{-1} u, \end{aligned} \tag{12}$$

where $G(\Delta, M) = W\,[(\tilde{M} + \tilde{M}^T)\circ\omega]\,W^T$ is twice the symmetrized gradient of $\mathrm{Tr}[M\,S(\Delta)]$, $g = W\,\mathrm{diag}(n_a) W^T$, $\tilde{M} = W^T M W$, $\omega_{ab} = (1-n_a-n_b)/(s_a+s_b)$, $s_a = \sqrt{n_a(1-n_a)+\varepsilon}$ and $\circ$ is the elementwise product. The remaining derivatives are

$$\alpha = \frac{\partial E}{\partial \nu_c} = 4V\big(R_v^T \rho^{vo} R_o + w^T C u\big), \qquad \Xi = \nabla_K E = -V\,\Pi, \tag{13}$$

and $H_{\rm off}$ has the off-diagonal blocks $H^{co} = -V(\bar{\Theta}u)R_o^T$, $H^{cv} = t_c R_c R_v^T - V(\bar{\Theta}w)R_v^T$, $H^{ov} = -V(2-\nu_c)R_o R_v^T$ (and their transposes), i.e. $\frac{1}{4}\partial E/\partial\rho^{IJ} = H^{IJ}$. The Jacobian of the constraints is also analytic. Gradients and Jacobian are verified against finite differences in Sec. 5.

Starting points. The problem is non-convex, so several random starts are run per parameter set (embedding states from random small Hamiltonians, X from a random symmetric matrix); a point is accepted if the constraint residual is $< 10^{-9}$, and the lowest accepted energy is kept. Seeding from the previous t_c can end in a poor local minimum (Sec. 6.1), so points are judged by their residuals and not by the energy alone.

$B_c = 5$. Random starts rarely converge at $B_c = 5$ (436 variables), so the search is seeded from a $B_c = 3$ solution with the two new ghosts asleep: $\Phi_c^{(5)} = \Phi_c^{(3)} \otimes |\mathrm{ghost}\rangle$ (one extra bath orbital filled, one empty) and $\rho^{(5)} = \rho^{(3)} \oplus \mathrm{diag}(0,1)$. This point is feasible and has the $B_c = 3$ energy; it is perturbed randomly and minimized. Sparse operators and caching make a $B_c = 5$ run take about 2 s.

Quasiparticle Hamiltonian and spectrum. The multipliers are recovered from the stationarity of $E + \sum_k \mu_k\kappa_k$ in Φ (least squares over the constraint components $\kappa_k = \rho^{II}_{ab} - {}^I\Delta_{ab}$; the residual of this fit is reported as the KKT residual), with $\lambda_{aa} = -\mu_{aa}/2$, $\lambda_{a<b} = -\mu_{ab}/4$. The quasiparticle Hamiltonian is $H_{qp} = H_{\rm off} - \mathrm{blockdiag}(\lambda_I)$, whose commutator with ρ vanishes at a solution (reported as $\|[H_{qp}, \rho]\|$). The hole Green's function of the core orbital is

$$G^h_{cc}(\omega) = \sum_{n \le N_{\rm occ}} \frac{|R_c^T \phi_n^{(c)}|^2}{\omega - \epsilon_n - i\eta}, \tag{14}$$

with ϵ_n, ϕ_n the eigenvalues (ascending) and eigenvectors of H_{qp} and $\phi_n^{(c)}$ the core block of ϕ_n; the poles ϵ_n are compared with $-E_i$ and the weights with p_i [Eq. (3)]. The total weight of G_{cc} is $R_c^T R_c$. The constraint Jacobian is rank deficient at solutions with pure ghost occupations (eigenvalues 0 or 1 of ${}^c\Delta$): some multiplier components are then undetermined (Sec. 6.2).

5 Checks

Gradients. The expressions of Sec. 4 were compared with central finite differences of Eq. (7) at random points (random ρ, R_I, ${}^I\Delta$, ν_c, K), for $B_c = 1, 3, 5$; maximum deviations:

Identity	max. deviation
$H^{IJ} = \frac{1}{4}\,\partial E/\partial \rho^{IJ}$	8×10^{-11}
$S_I h_I = \frac{1}{4}\,\partial E/\partial R_I$	1×10^{-10}
$G(\Delta, M) = 2\nabla^s \mathrm{Tr}[MS(\Delta)]$	4×10^{-10}
$G({}^c\Delta, h_c R_c^T) - \Lambda_c$ vs. $\frac{1}{2}\nabla^s_{{}^c\Delta} E$ (fixed R_c)	6×10^{-10}
$\alpha = \partial E/\partial \nu_c$	2×10^{-10}
$\Xi = \nabla_K E$	2×10^{-10}

The complete gradient of the minimization (with respect to x_I and X) agrees with finite differences with a relative error that decreases with the step down to 4×10^{-7} (limited by the stiffness of S^{-1} for nearly pure ${}^I\Delta$), and the constraint Jacobian to 3×10^{-10}.

Limits. For $B_c = 1$ and $B_c = 3$:

V/g	t_c/g	$E - E_{\mathrm{ref}}$ ($B_c = 1$)	$E - E_{\mathrm{ref}}$ ($B_c = 3$)	ν_c
0	0.5	$+4.4 \times 10^{-8}$	$+4.4 \times 10^{-8}$	1.995893
0	0.1	$+4.4 \times 10^{-8}$	$+4.4 \times 10^{-8}$	1.999835
1	0	$+5 \times 10^{-14}$	$+3 \times 10^{-14}$	2
2	0	$+3 \times 10^{-14}$	$+3 \times 10^{-13}$	2

At $V = 0$ the reference is the exact non-interacting energy (two lowest levels of the 3×3 one-body problem, times two spins; the 4×10^{-8} is the effect of $\varepsilon = 10^{-9}$); at $t_c = 0$ it is $-2E_c - 2g$. At $t_c = 0$ the energy and $\nu_c = 2$ are exact for all V, but R_c is not determined by the stationarity conditions (E does not depend on R_c at $t_c = 0$), so G_{cc} carries no information: V only enters through $(2 - \nu_c)$ and u, w, which vanish for the product state.

Quality of the solutions. Throughout Sec. 6 the accepted $B_c = 3$ and $B_c = 5$ points have constraint residual $\lesssim 10^{-9}$ (typically 10^{-14}) and KKT residual $\leq 10^{-6}$ unless stated otherwise.

6 Results for $B_c = 1$, 3 and 5

$B_c = 3$ and $B_c = 5$ mean two and four ghost orbitals on the core, $B_c = 1$ none (standard Gutzwiller). The $B_c = 3$ and $B_c = 5$ results are from the direct minimization above ($B_c = 5$ seeded from the $B_c = 3$ solution, plus random starts). The $B_c = 1$ results come from converged calculations of the same problem; at $(V, t_c) = (1, 0.1)$ the direct minimization reproduces them to eight digits in E (and to the printed precision, 10^{-3}, in the pole).

6.1 Energy and ν_c

Figure 1 shows $E_{\mathrm{gGut}} - E_{\mathrm{ED}}$ (left), the energy gained by the ghosts (middle) and $|\nu_c - \nu_c^{\mathrm{ED}}|$ (right) for $B_c = 1$, 3 and 5 versus t_c, for $V/g = 0.5, 1, 2$:

- $B_c = 1$**:** $E_{\mathrm{gGut}} - E_{\mathrm{ED}} \simeq 1.3 \times 10^{-3}\, V^2 t_c^2$ over the grid $t_c \in [0.025, 0.5]$ (fitted exponent in t_c: 1.94–1.99; $\Delta E/(V^2 t_c^2)$ between 1.25 and 1.55×10^{-3}), and $|\nu_c - \nu_c^{\mathrm{ED}}| \propto t_c^2$ as well.

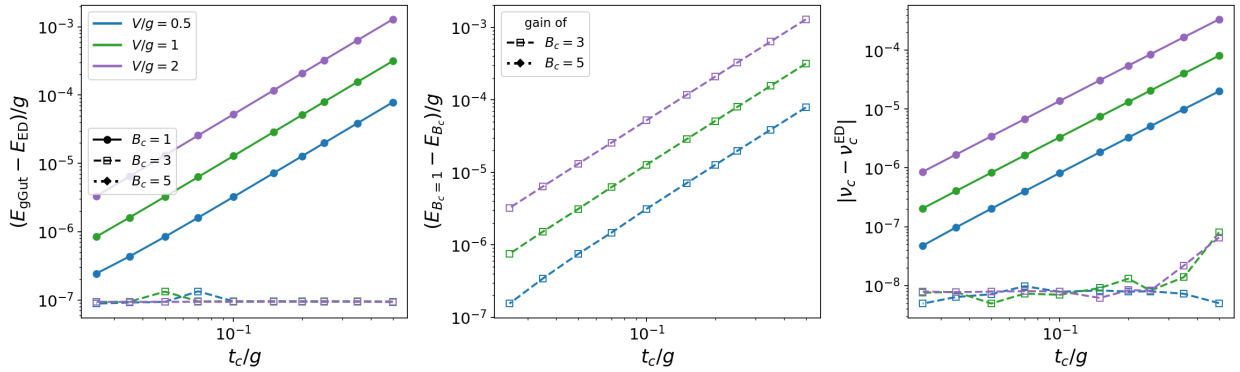


Figure 1: Left: $E_{\mathrm{gGut}} - E_{\mathrm{ED}}$ for $B_c = 1$ (full lines), $B_c = 3$ (dashed, open squares) and $B_c = 5$ (dotted, diamonds); colour is V. Middle: energy gained by the ghosts, $E_{B_c=1} - E_{B_c}$, for $B_c = 3$ and $B_c = 5$ (the two coincide). Right: $|\nu_c - \nu_c^{\mathrm{ED}}|$. Log–log axes; $E_c/g = 10$.

- $B_c = 3$**:** $E_{\mathrm{gGut}} - E_{\mathrm{ED}} \simeq 9.4 \times 10^{-8}$ for all (V, t_c), flat in both, and $|\nu_c - \nu_c^{\mathrm{ED}}|$ at 10^{-8} (up to 8×10^{-8} at $t_c/g = 0.5$). This floor is set by the regularization shift (2.6×10^{-8} at $\varepsilon = 10^{-12}$), not by the ansatz.
- $B_c = 5$**:** at all 30 points (V, t_c) of the grid the converged $B_c = 5$ minimum has $E_{\mathrm{gGut}} - E_{\mathrm{ED}}$ between 8.8 and 9.6×10^{-8} and $E_{B_c=5} - E_{B_c=3}$ between -2×10^{-10} and $+4 \times 10^{-11}$; $|\nu_c - \nu_c^{\mathrm{ED}}| \leq 1.8 \times 10^{-8}$ and the KKT residual is $\leq 5.5 \times 10^{-6}$. Per point, 4–9 of 14 runs (8 seeded, 6 random) converged, and all converged runs agree to $\leq 10^{-9}$ in energy; of the 180 random $B_c = 5$ starts, 8 converged, to the same energy. The two extra ghosts lower the energy by nothing.
- The ghosts therefore recover almost all of the $B_c = 1$ error: the energy gain $E_{B_c=1} - E_{B_c=3}$ is 64–99.99% of it ($\geq 97\%$ for $V/g = 2$, $\geq 89\%$ for $V/g = 1$; the smallest fractions are at the smallest t_c, where the shift floor is comparable to the error).

Starting each t_c from the previous solution at $V/g = 1$ ended in a poor local minimum at six values of t_c ($E - E_{\mathrm{ED}}$ up to 7×10^{-2}, KKT up to 10^{-2}); these points were recomputed from 4–13 random starts each and now lie at 9.4–9.5 $\times 10^{-8}$ with KKT $\leq 10^{-6}$.

6.2 Hole spectral function

Dependence on B_c. Figure 2 compares the exact hole spectral function with the gGut ones, and Tab. 3 lists the dominant poles. The $B_c = 1$ pole sits at -10.000 for *every* V: it is the bare core level. At $B_c = 3$ the single pole moves with V (-9.97, -9.87, -9.50 for $V/g = 0.5, 1, 2$) in the direction of the exact main pole (-9.76, -9.17, -7.53 at $t_c/g = 0.05$), but covers only 15–20% of the exact displacement from $-E_c$ (0.035 of 0.236, 0.131 of 0.828, 0.499 of 2.47), and its weight (0.997, 0.988, 0.955) stays far above the exact one (0.90, 0.73, 0.52). The ghosts bring a V dependence into the pole, but no satellites. Going from $B_c = 3$ to $B_c = 5$ changes nothing visible: the $B_c = 5$ pole and weight coincide with the $B_c = 3$ ones to the printed digits (Tab. 3).

Why no satellites (interpretation; $B_c = 3$ and $B_c = 5$). Of the occupied quasiparticle levels, one is the o-like level, one carries the main pole, and the remaining ones are ghost levels with essentially no core weight, $|R_c^T \phi^{(c)}|^2 < 10^{-6}$ for the one ghost level at $B_c = 3$ and $\leq 5 \times 10^{-6}$

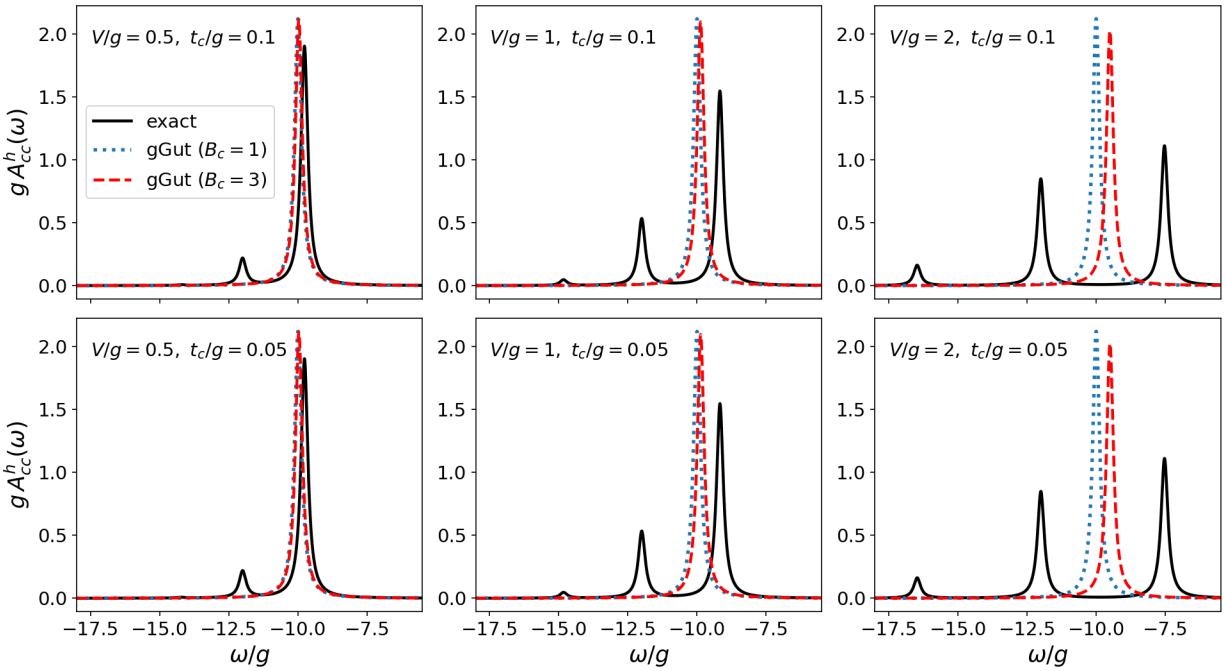


Figure 2: Hole spectral function $A^h_{cc}(\omega) = \sum_n w_n\,(\eta/\pi)/[(\omega-\omega_n)^2+\eta^2]$ with $\eta = 0.15$: exact (solid), gGut with $B_c = 1$ (blue dotted), $B_c = 3$ (red dashed) and $B_c = 5$ (green dotted; it lies on top of the $B_c = 3$ curve). Rows: $t_c/g = 0.1$ and 0.05; columns: $V/g = 0.5, 1, 2$ ($E_c/g = 10$).

		ED main pole		gGut $B_c = 1$		gGut $B_c = 3$		gGut $B_c = 5$	
V/g	t_c/g	position	weight	position	weight	position	weight	position	weight
0.5	0.1	−9.7649	0.8972	−10.001	0.9999	−9.968	0.9969	−9.968	0.9969
0.5	0.05	−9.7642	0.8972	−10.000	0.9999	−9.965	0.9968	−9.965	0.9968
1	0.1	−9.1728	0.7287	−10.001	0.9999	−9.872	0.9882	−9.872	0.9882
1	0.05	−9.1719	0.7286	−10.000	0.9999	−9.869	0.9881	−9.869	0.9881
2	0.1	−7.5295	0.5238	−10.000	0.9999	−9.503	0.9547	−9.503	0.9547
2	0.05	−7.5283	0.5237	−10.000	0.9999	−9.501	0.9546	−9.501	0.9546

Table 3: Dominant hole pole of G_{cc}. ED additionally has satellites at about −12.0 (total weight 0.10, 0.25, 0.40 for $V/g = 0.5, 1, 2$, split into two lines $\sim 10^{-4}$ apart) and at −14.2, −14.8, −16.5 (weights 0.003, 0.02, 0.08); gGut has none (all other occupied levels have weight $< 4\times10^{-6}$ at $B_c = 1$, $< 2\times10^{-4}$ at $B_c = 3$ and $\le 5\times10^{-6}$ at $B_c = 5$).

for the three at $B_c = 5$ (Tab. 3). The constraint Jacobian is rank deficient, so some multiplier components are undetermined, and they only move these weightless levels: rank 7 of 8 at $B_c = 3$ and 13 of 17 at $B_c = 5$ (four singular values $\lesssim 2\times10^{-7}$ against $\ge 3.7\times10^{-4}$ for the rest; $V/g = 1$, $t_c/g = 0.1$). At $B_c = 3$ two minima with the same energy put the ghost level at −11.27 and −2.81 ($V/g = 1$, $t_c/g = 0.5$), and at $B_c = 5$ two runs give $(-4.99, -4.34, -2.57)$ and $(-2.40, -2.27, -1.37)$ ($V/g = 1$, $t_c/g = 0.1$); in each pair the main pole and its weight are identical (−9.892, 0.98635 and −9.872, 0.9882). The ghosts appear to carry the correlation through ν_c and K, i.e. through the terms $-V(\bar\Theta u)R_o^T$ and $-V(\bar\Theta w)R_v^T$ of H_{qp}, but not through R_c, which is all that Eq. (14) sees. This is an interpretation of the numbers, not a proof. Counting: $B_c = 3$ has only two occupied core-like levels, so it could give at most two hole poles, whereas $B_c = 5$ has room for three; the

extra ghosts nevertheless carry no weight.

7 Conclusions

1. **Main result: gGut does not reproduce the satellite structure of the model**, which was the objective of the test. For $B_c = 1$, 3 and 5, $V/g = 0.5, 1, 2$ and $t_c/g = 0.05$–0.5, the hole spectral function from the quasiparticle Hamiltonian, Eq. (14), has a single pole with weight 0.95–0.9999, whereas the exact spectrum has a main pole and satellites near -12 and -14 to -16.5 that carry 10%, 27% and 48% of the weight at $V/g = 0.5, 1, 2$ (Fig. 2, Tab. 3). The main pole itself is displaced from the exact one (-9.87 vs. -9.17 at $V/g = 1$) and, in the strongest coupling, by 2 (-9.50 vs. -7.53 at $V/g = 2$). Adding ghosts improves the ground state (below, Fig. 1) but does not produce satellites.

2. The Hamiltonian is Eq. (1) with a core–virtual hopping; the core–occupied hopping would not change the ground state ($|220\rangle$ stays exact), so it cannot fix the core quantities of the gGut problem.

3. The energy functional (7) and its analytic gradients are verified (finite differences to $\sim 10^{-10}$; the $V = 0$ and $t_c = 0$ limits).

4. Direct minimization with $B_c = 3$ (and $B_c = 5$, below) finds stationary points of problem (9) with $E - E_{\mathrm{ED}} \approx 10^{-7}$ (vs. $\approx 1.3 \times 10^{-3} V^2 t_c^2$ at $B_c = 1$) and ν_c equal to the exact one to $\sim 10^{-8}$: the ghosts improve the ground state.

5. $B_c = 5$ (four ghosts): the minimizer, seeded from the $B_c = 3$ solution and from random starts, returns the $B_c = 3$ energy, ν_c and spectrum at all 30 grid points; the extra ghosts stay asleep.

6. At $B_c = 1$ the single pole is the bare core level for all V; at $B_c = 3$ and 5 it moves with V but covers only 15–20% of the exact displacement, and the ghost levels carry no core weight.